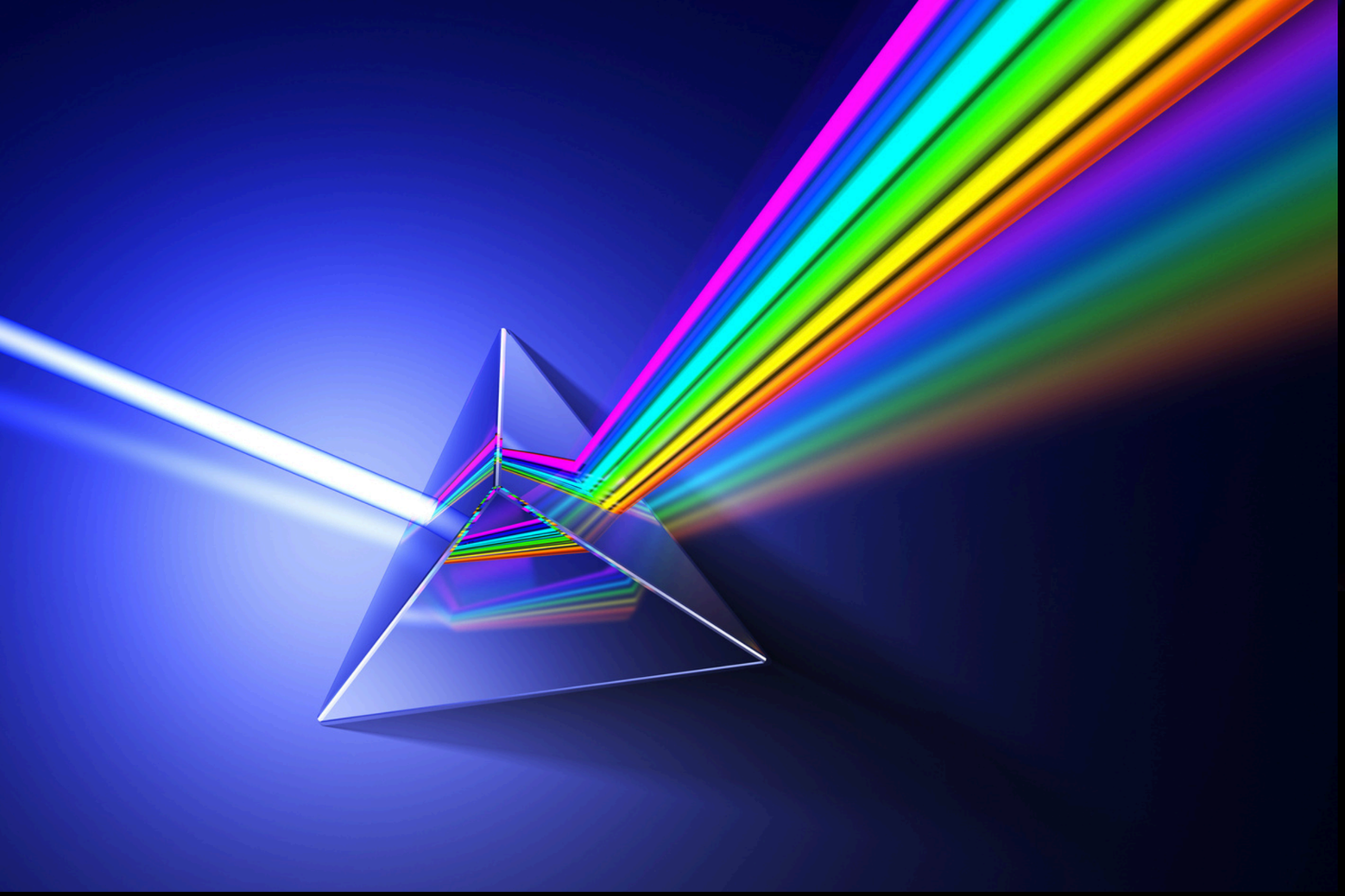
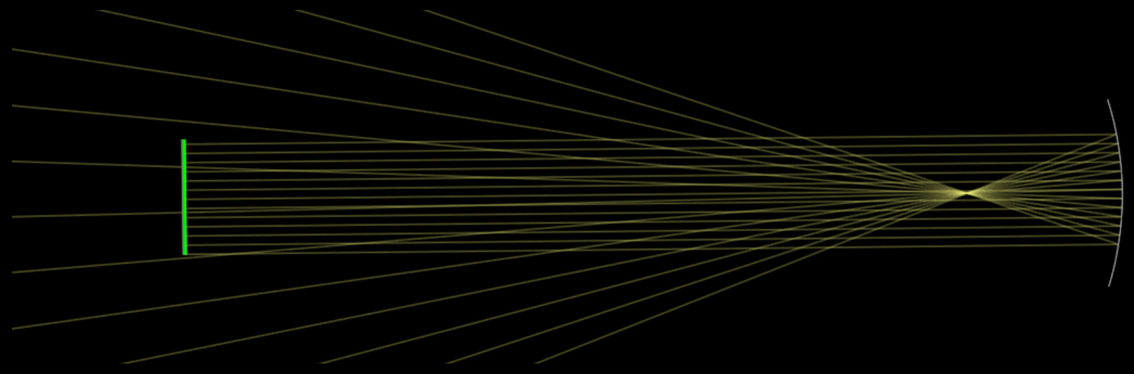
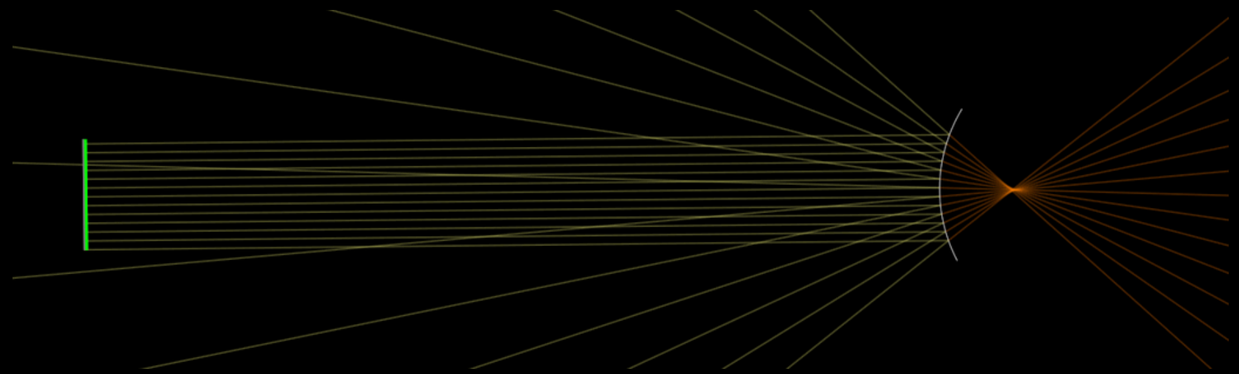
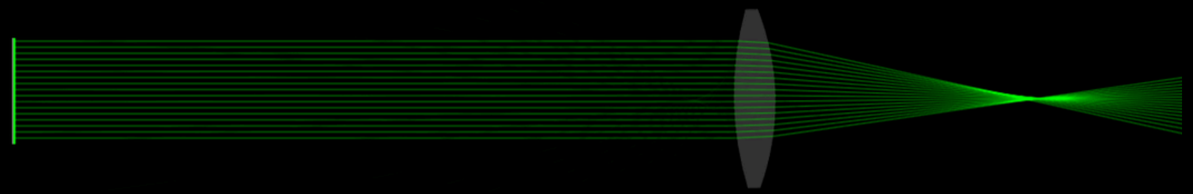


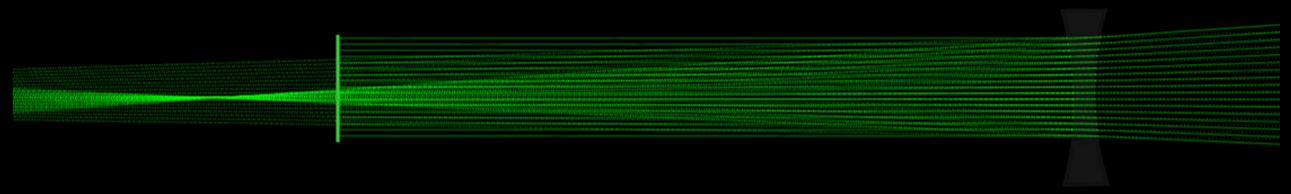
# Optics



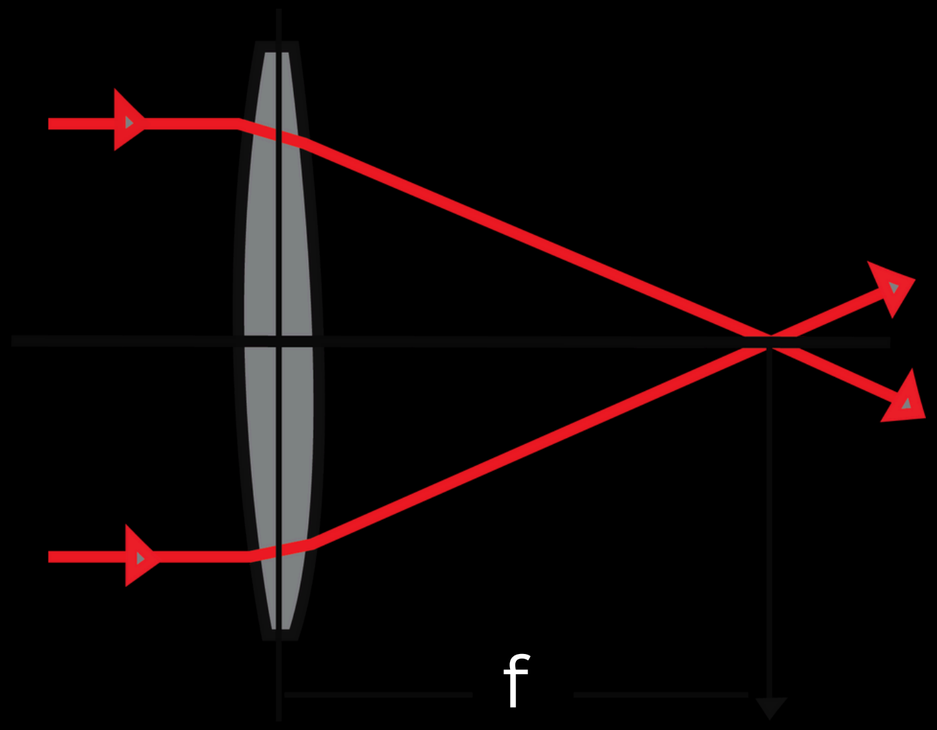


# Light Ray Diagram

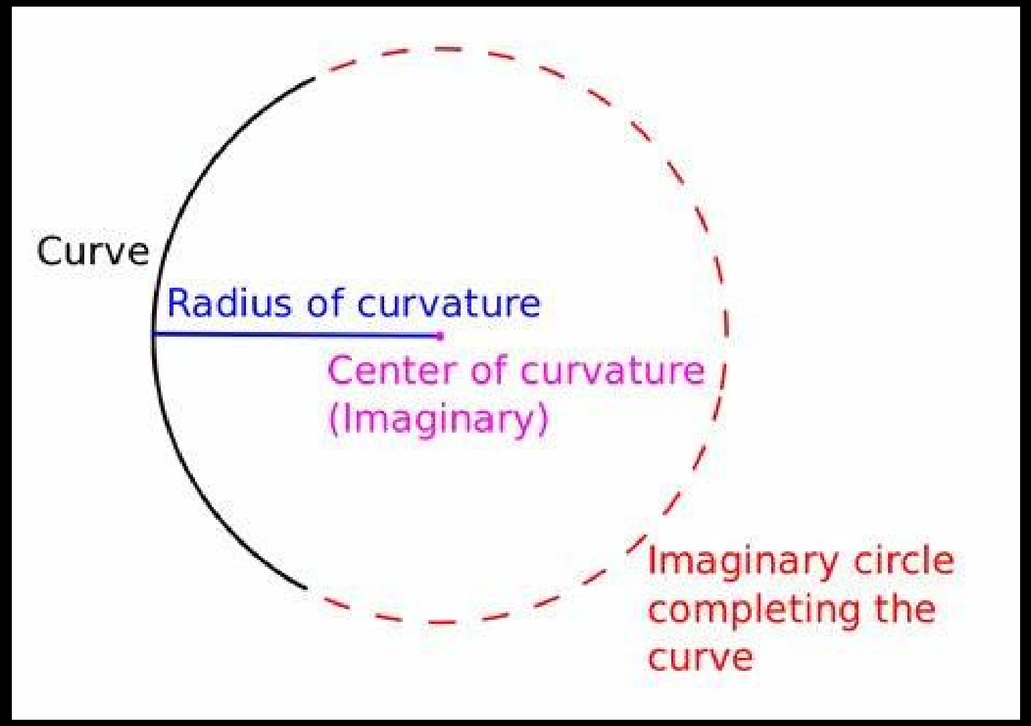




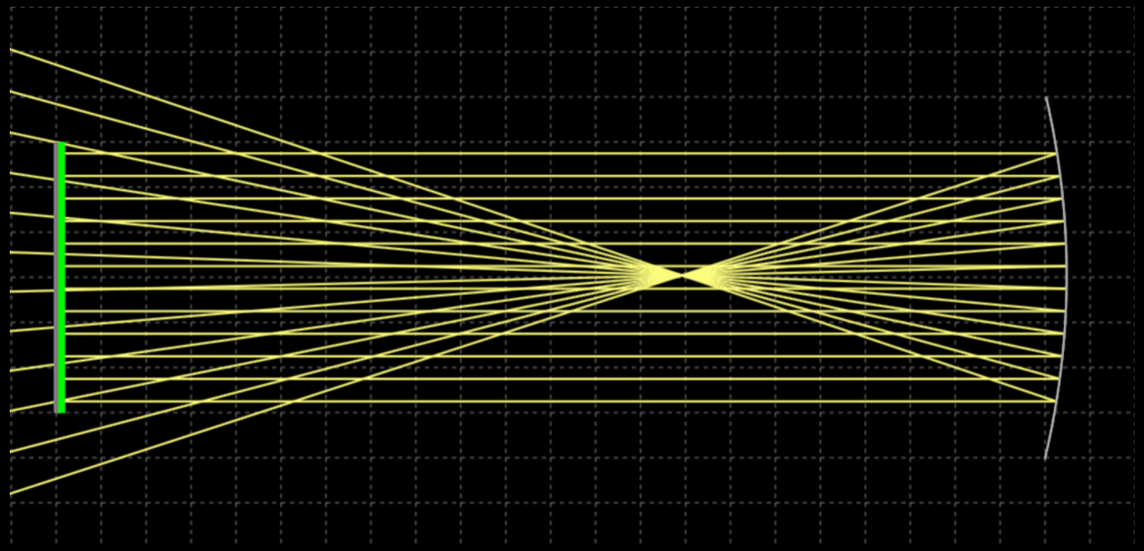
## Focal Length



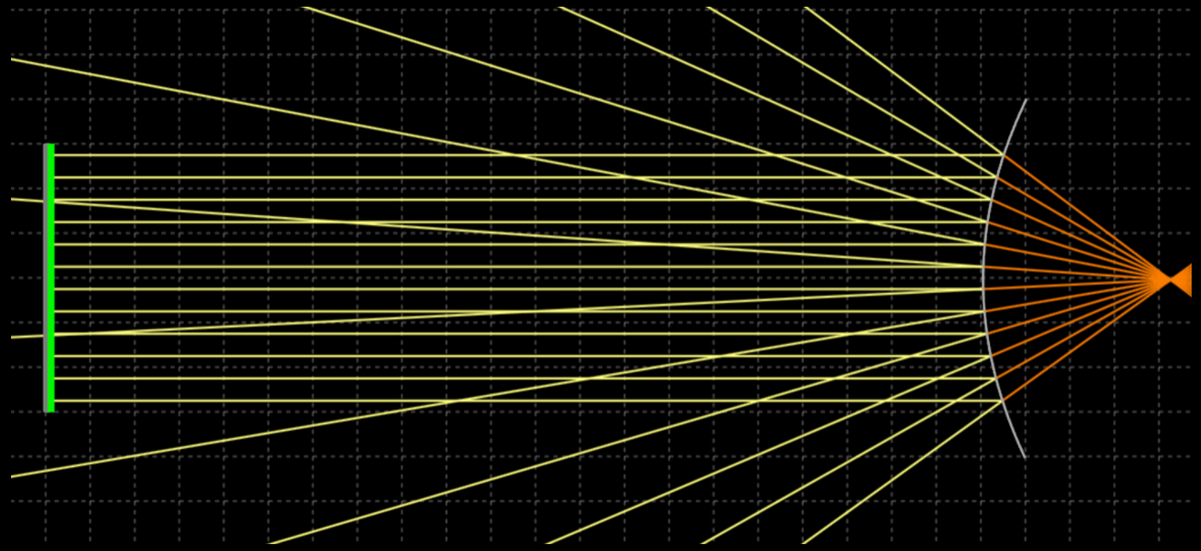

## Radius of Curvature



# Concave & Convex Mirrors

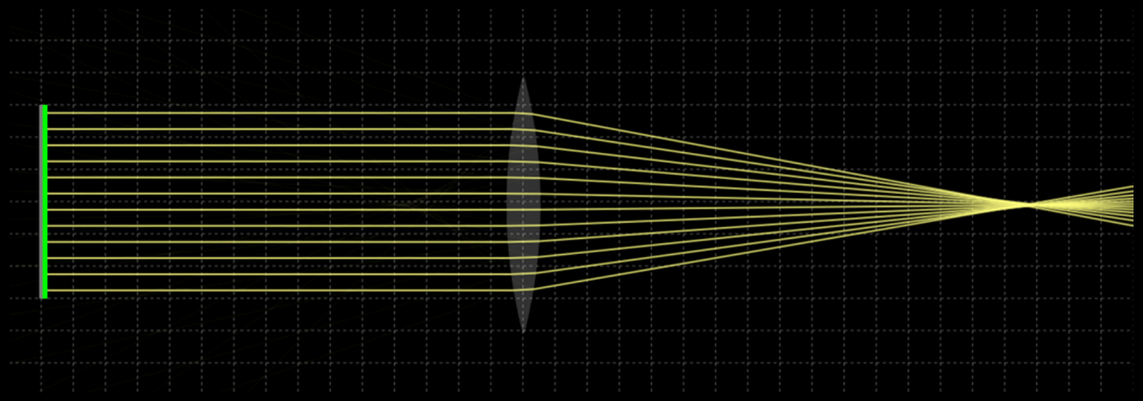


Concave Mirror

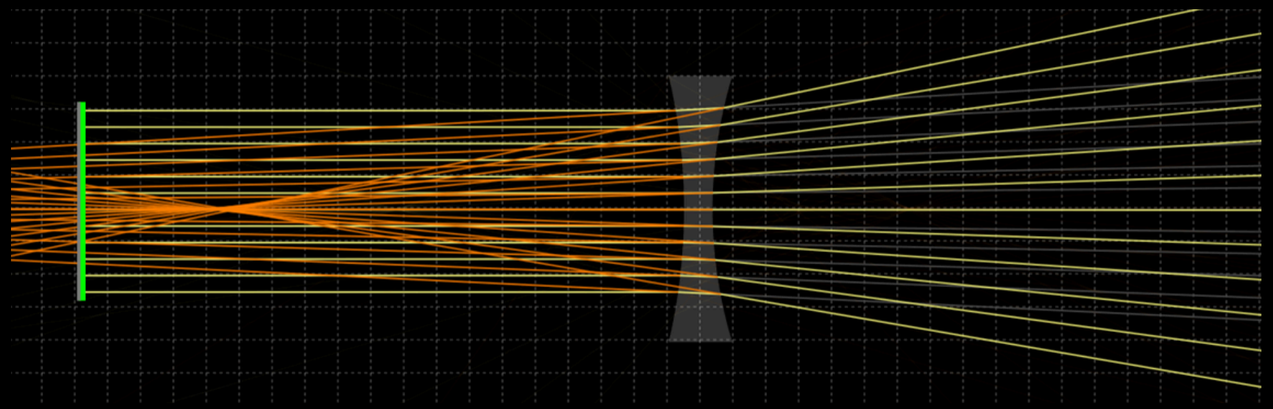


Convex Mirror

# Concave & Convex Lens

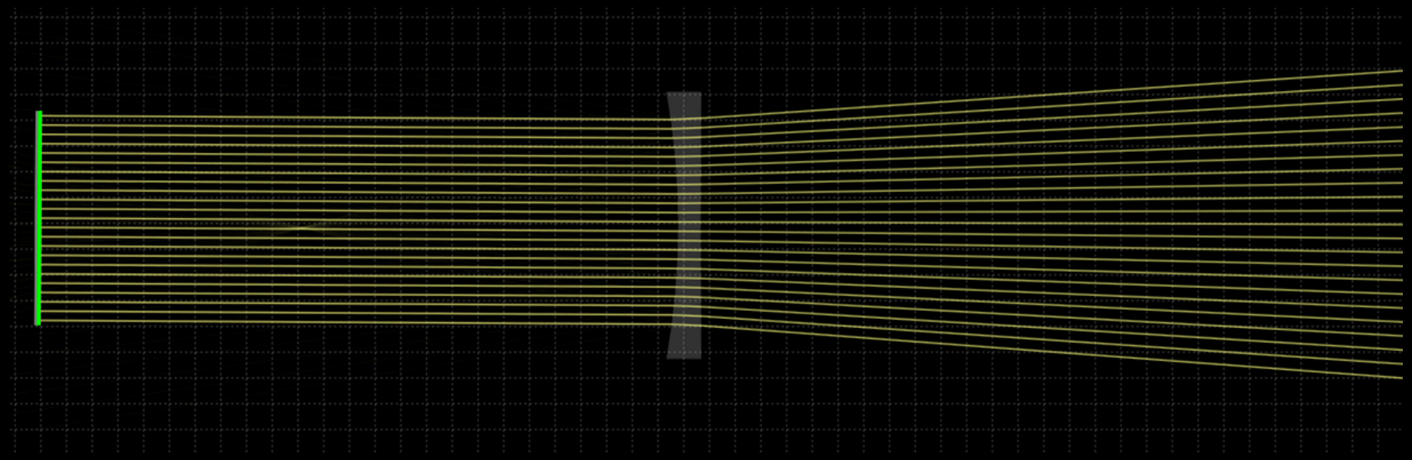


Convex Lens



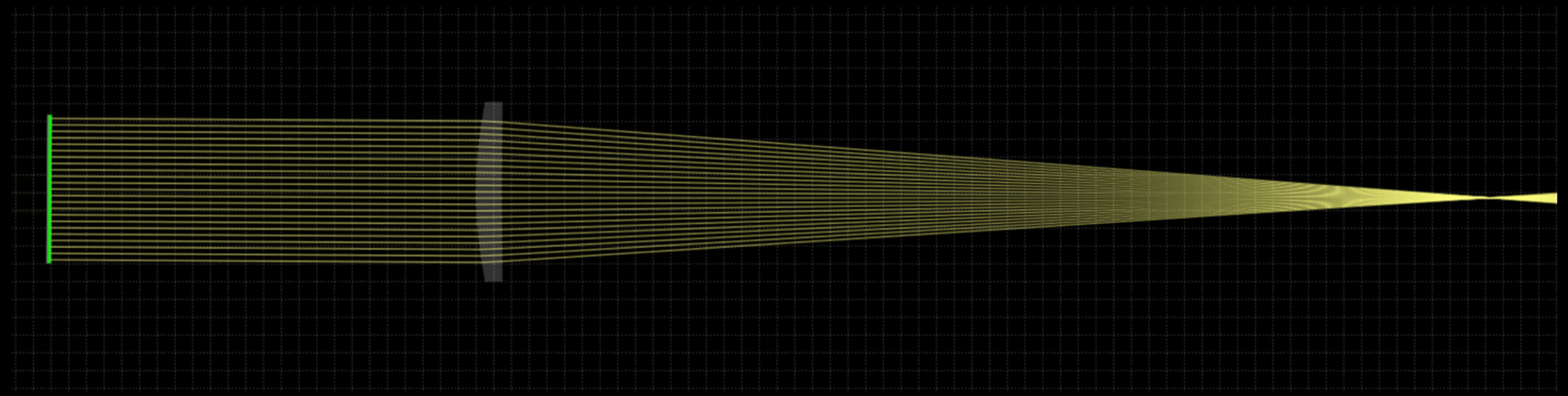
Concave Lens

# Special Lenses

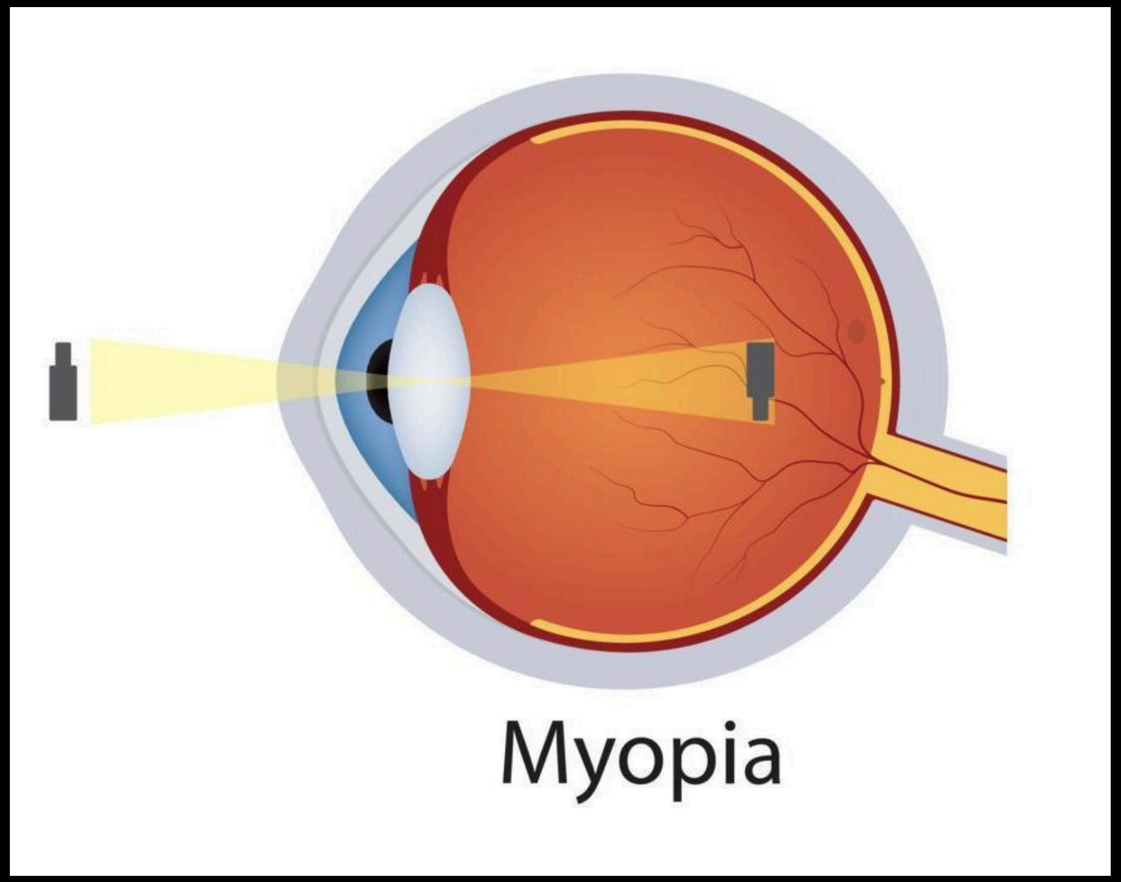


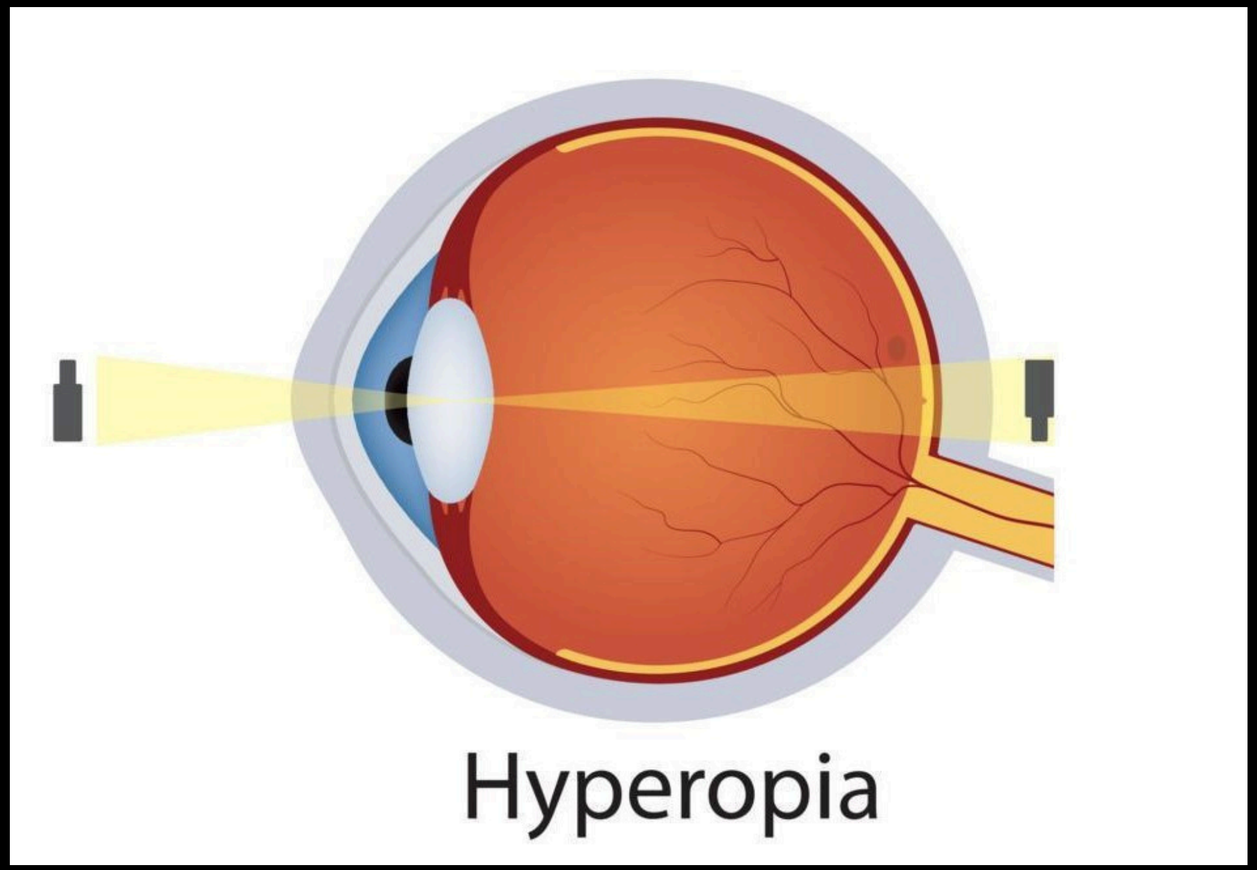
Plano-Concave Lens

Plano-Convex Lens

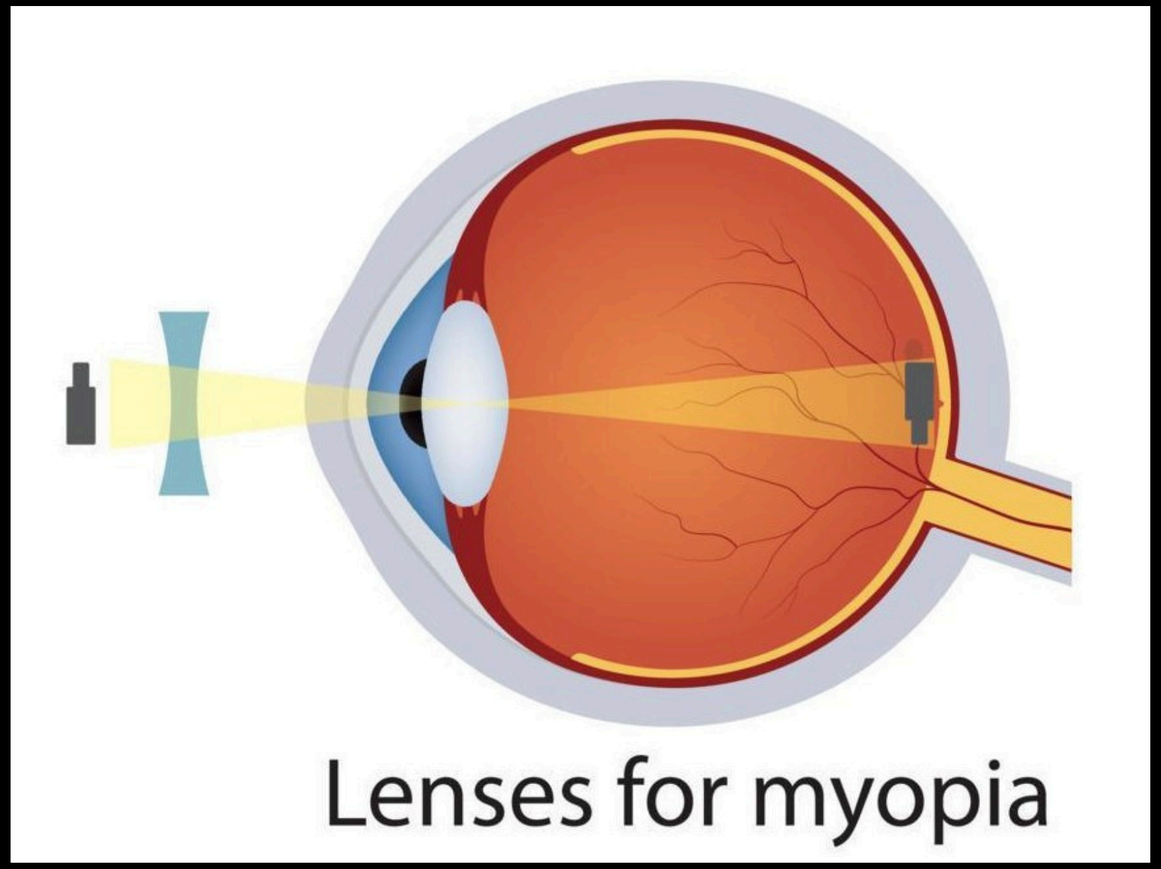


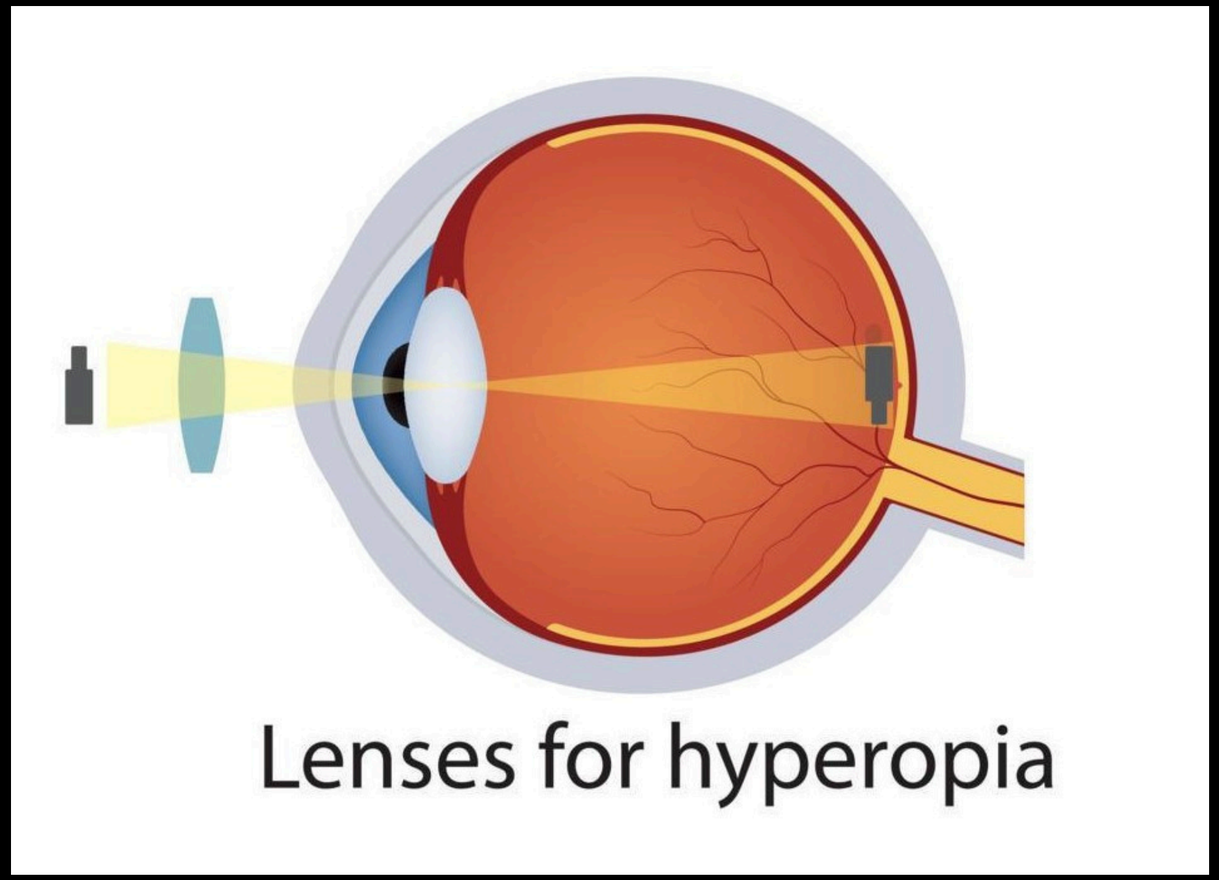
# Eye Conditions



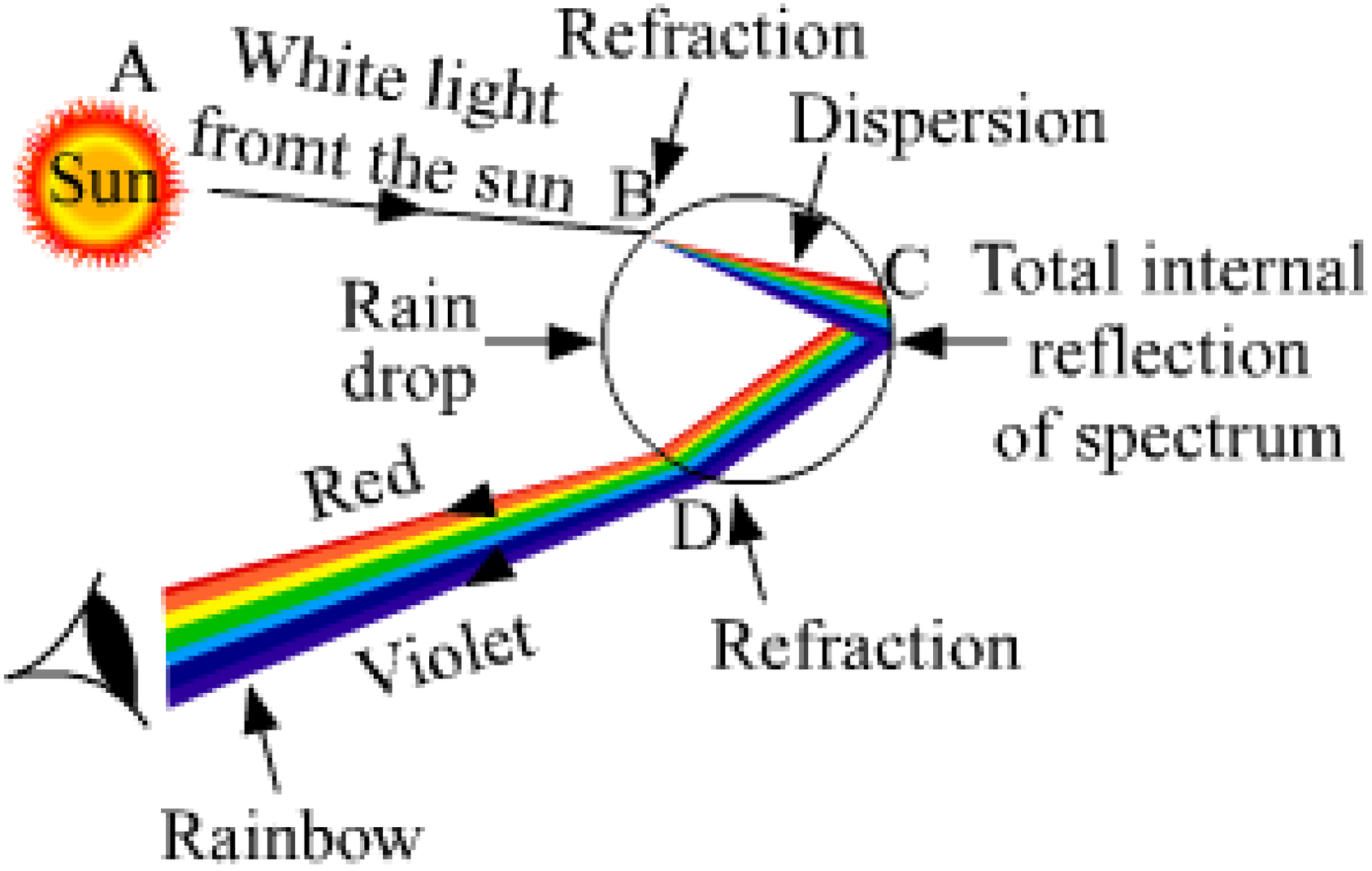



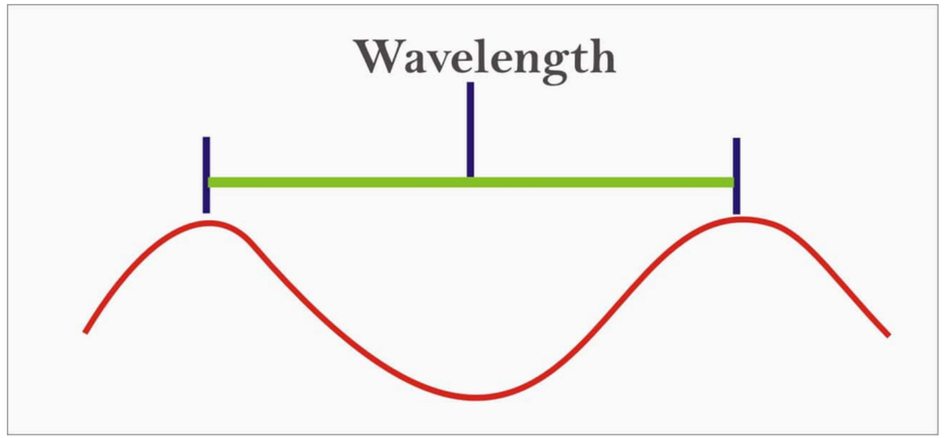
# Eye Conditions - Corrections



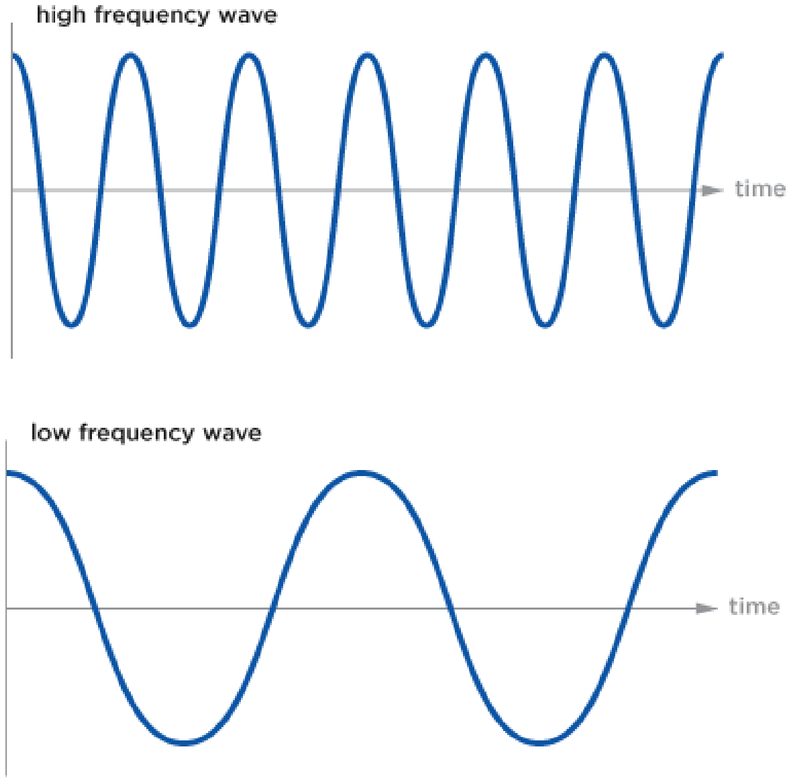
# Critical Angle
# &
# Total Internal Reflection


A
Sun
White light
fromt the sun
Refraction
B
Dispersion
C
Total internal
reflection
of spectrum
Rain
drop
Red
D
Violet
Refraction
Rainbow

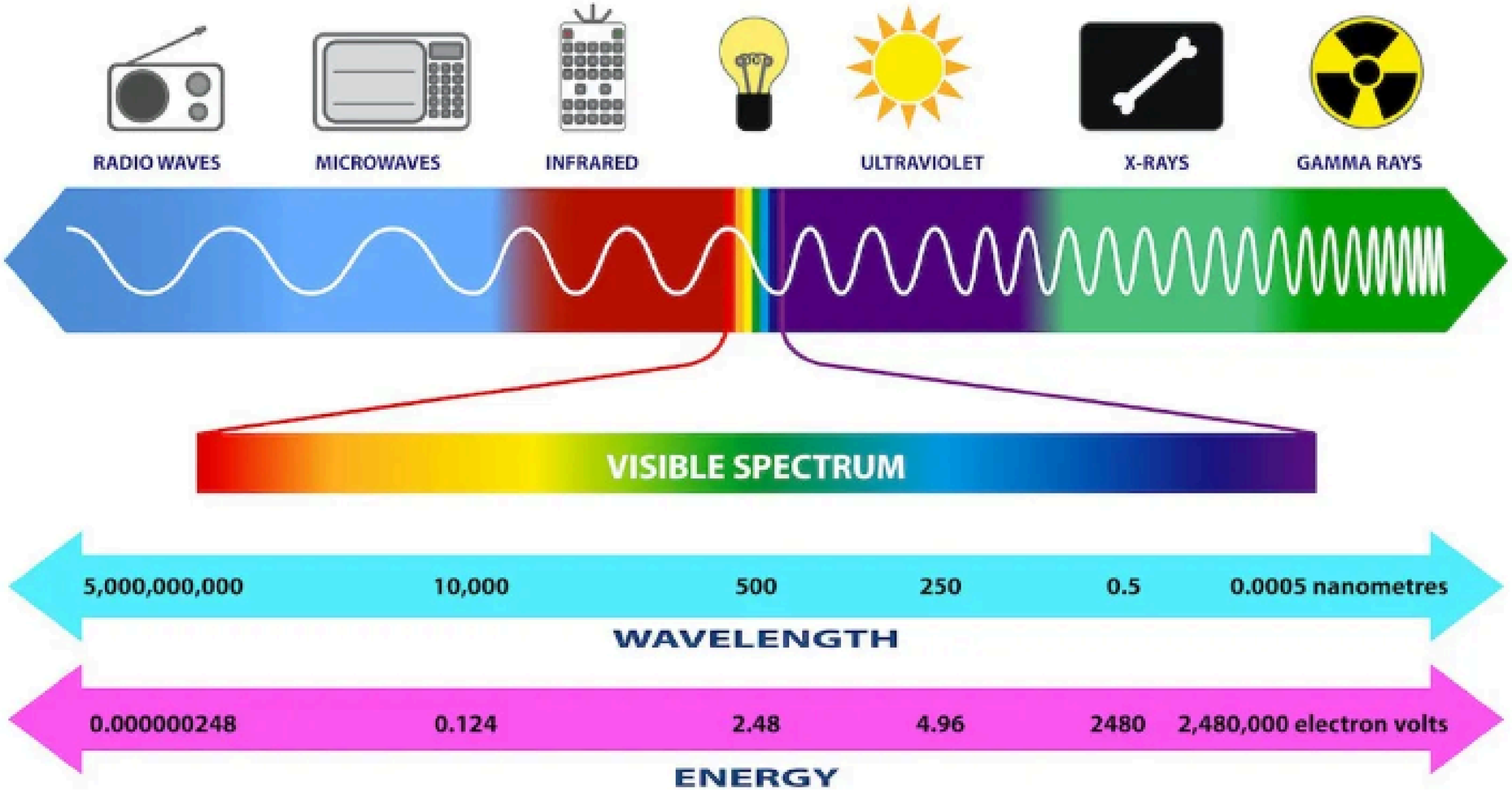



Wavelength

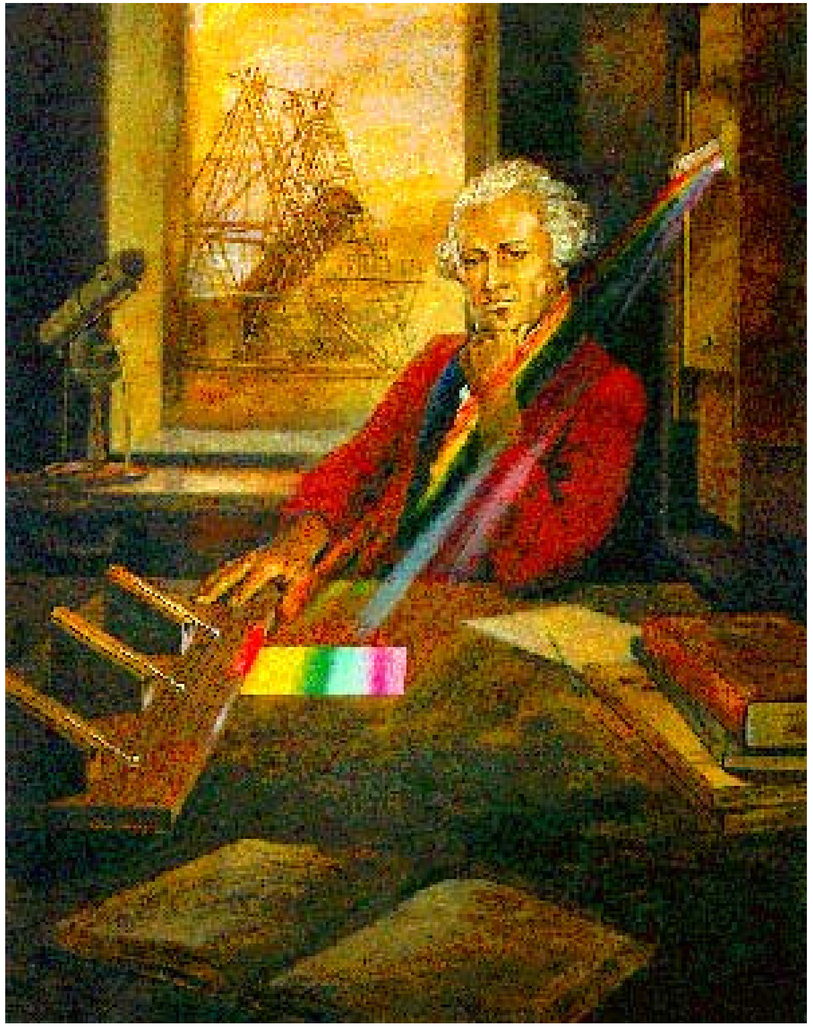
Frequency



# ELECTROMAGNETIC SPECTRUM

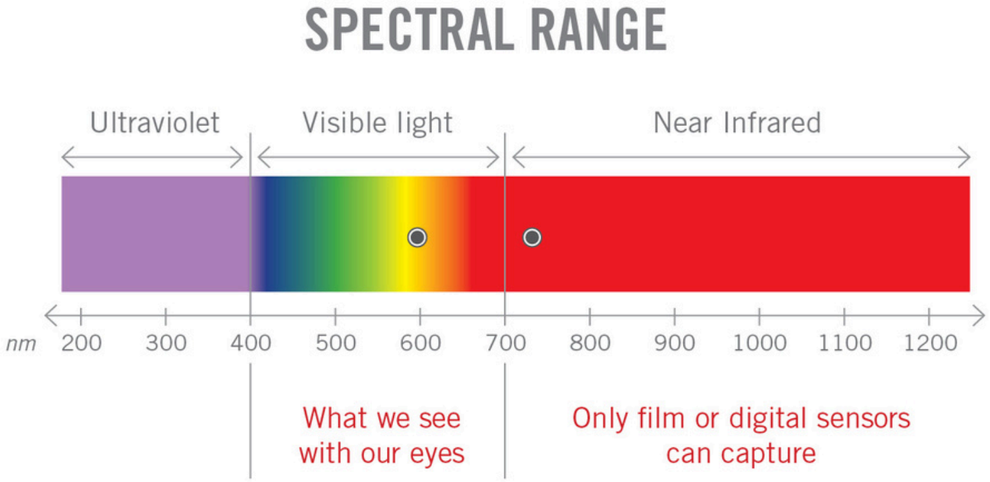


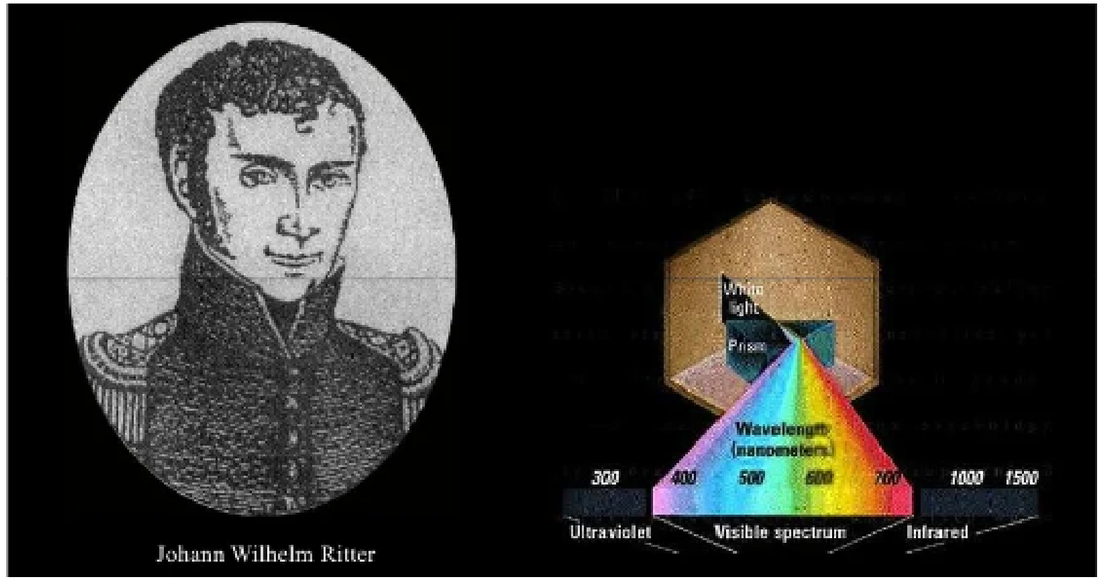
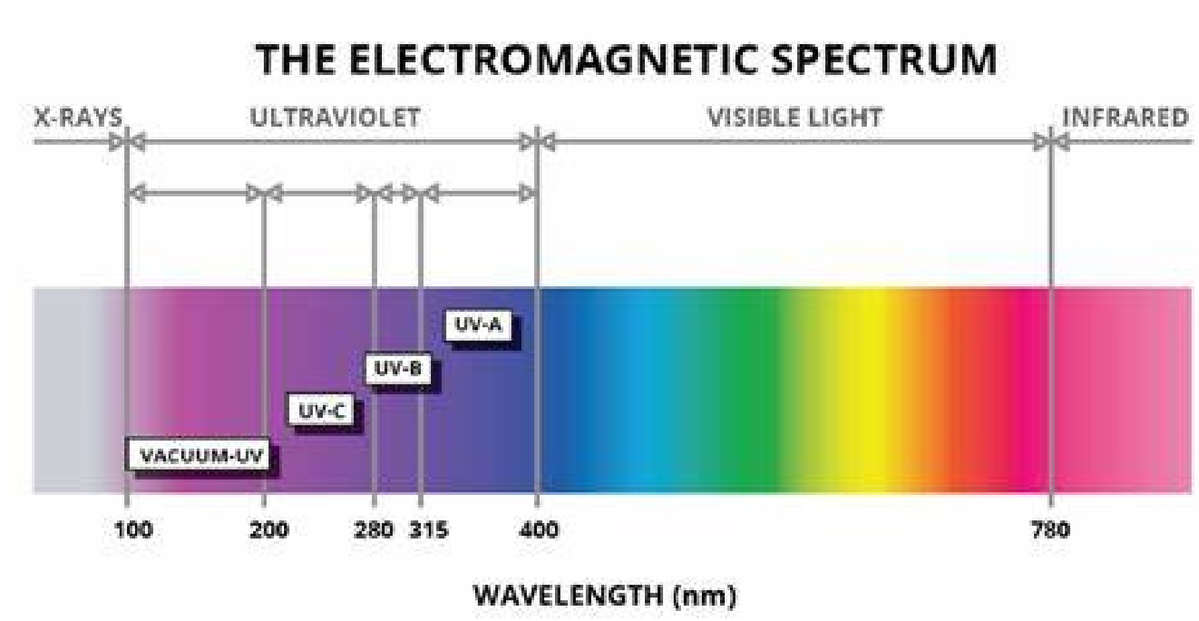
# Discovery of UV and IR Spectrum



## **Infra Red Spectrum** *(Calorific Waves)*

- Discovered by Willaim Herschel
- Used Simple Thermometeres for the Detection.
- Discovered in the year 1800



# **UV Spectrum** (*actinic rays)*

- Discovered by Johann Wilhelm Ritter
- Used Silver Chloride Chemical
- Discovered in the year 1801